# Module 0

## Installation and Setup

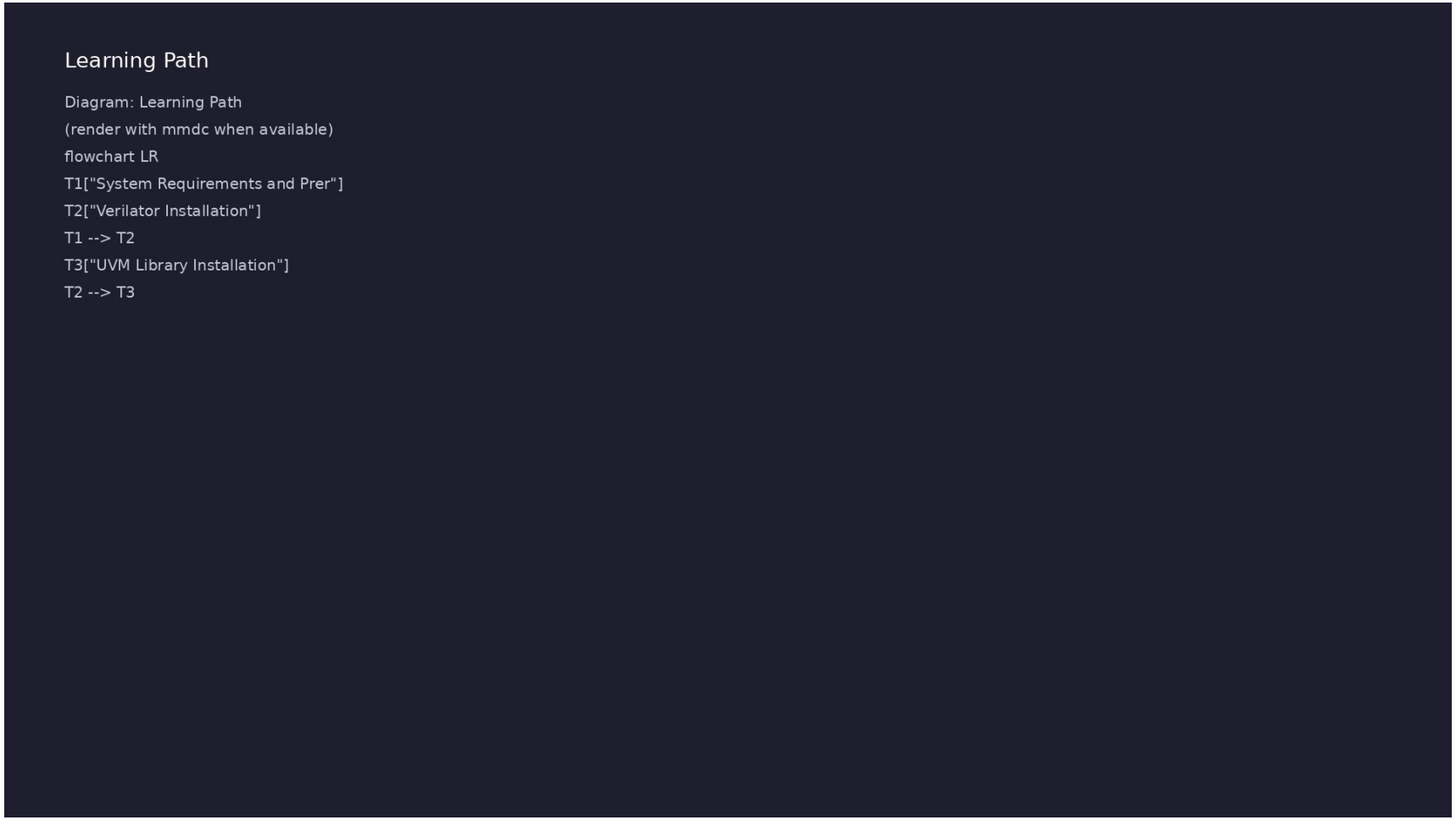
# Learning objectives

- Set up development environment and verify installation

# Prerequisites

- See module README

# Learning path




Set up development environment and verify installation

# Overview

- This module covers the complete setup of your verification environment, including all required tool…

# Syllabus topics

# 1. System Requirements and Prerequisites (1/3)

- Operating System Support
- Linux (Ubuntu/Debian, CentOS/RHEL, Fedora)
- macOS (Intel and Apple Silicon)
- Windows (WSL2 recommended)
- Hardware Requirements
- Minimum 8GB RAM (16GB+ recommended)

# 1. System Requirements and Prerequisites (2/3)

- 10GB free disk space
- Multi-core processor recommended
- Software Prerequisites
- C/C++ compiler (GCC 7+, Clang 10+, or MSVC)
- Make or Ninja build system
- Git

# 1. System Requirements and Prerequisites (3/3)

- Perl (for UVM library)

# 2. Verilator Installation (1/6)

- What is Verilator?
- Open-source Verilog/SystemVerilog simulator
- Fast compilation and simulation
- Limited SystemVerilog support (see limitations section)
- Automated Installation (Recommended)
- Using the installation script:

# 2. Verilator Installation (2/6)

- The script automatically:
- Checks for existing installations
- Installs system dependencies (build tools, libraries)
- Sets up git submodule in `tools/verilator/`
- Builds and installs Verilator
- Verifies the installation

# 2. Verilator Installation (3/6)

- Manual Installation Methods
- Linux Installation
- Ubuntu/Debian: `sudo apt-get install verilator`
- CentOS/RHEL: `sudo yum install verilator` or `sudo dnf install verilator`
- Fedora: `sudo dnf install verilator`
- Building from source (latest features)

# 2. Verilator Installation (4/6)

- macOS Installation
- Homebrew installation: `brew install verilator`
- MacPorts installation (alternative)
- Building from source
- Windows/WSL2 Installation
- Installing in WSL2 Ubuntu: `sudo apt-get install verilator`

# 2. Verilator Installation (5/6)

- Building from source in WSL2
- Verifying installation
- Uninstallation
- Verification Steps
- Check Verilator version: `verilator --version`
- Run simple Verilator test

# 2. Verilator Installation (6/6)

- Verify compilation works

# 3. UVM Library Installation (1/5)

- What is UVM?
- Universal Verification Methodology
- IEEE 1800.2-2017 standard
- Industry-standard verification framework
- Automated Installation (Recommended)
- Using the installation script:

# 3. UVM Library Installation (2/5)

- The script automatically:
- Downloads Accellera UVM 2017-1.0 tarball to
  `tools/uvm-2017/`
- Extracts and sets up UVM library
- Creates environment setup script
  (`scripts/setup_uvm_env.sh`)
- Configures UVM library paths
- Manual Installation Methods

# 3. UVM Library Installation (3/5)

- From Accellera Tarball:
- From System Package (if available):
- Some Linux distributions provide UVM packages
- Check your distribution's package manager
- Uninstallation
- Environment Variables

# 3. UVM Library Installation (4/5)

- `UVM_HOME`: Path to UVM library
- `UVM_VERBOSITY`: Default verbosity level
- Include paths for compilation
- Verification Steps
- Check UVM library exists: `ls $UVM_HOME/uvm_pkg.sv`
  (after sourcing `scripts/setup_uvm_env.sh`)
- Compile simple UVM test

# 3. UVM Library Installation (5/5)

- Verify UVM classes available

# 4. Commercial Simulator Setup (Optional) (1/2)

- Synopsys VCS
- Installation from Synopsys
- License setup
- UVM library integration
- Mentor QuestaSim/ModelSim
- Installation from Siemens

# 4. Commercial Simulator Setup (Optional) (2/2)

- License setup
- UVM library integration
- Cadence Xcelium
- Installation from Cadence
- License setup
- UVM library integration

# 5. Build System Configuration (1/3)

- Makefile Setup
- Basic Makefile for Verilator
- Compilation flags
- Include paths
- Test execution
- Verilator Compilation

# 5. Build System Configuration (2/3)

- SystemVerilog flags: `-sv`, `--timing`
- Include paths: `+incdir+$UVM_HOME` (UVM_HOME already points to the src directory)
- Optimization flags
- Coverage options
- Commercial Simulator Setup
- Compilation scripts

# 5. Build System Configuration (3/3)

- Simulation scripts
- Waveform generation

# 6. IDE Setup and Configuration (1/3)

- Recommended IDEs
- VS Code with SystemVerilog extension
- Verible (SystemVerilog linter/formatter)
- Sigasi Studio (commercial)
- Vim/Neovim with LSP
- VS Code Configuration

# 6. IDE Setup and Configuration (2/3)

- SystemVerilog extension setup
- Verilator integration
- Debugging configuration
- Task runner setup for simulations
- Extension recommendations
- Editor Configuration

# 6. IDE Setup and Configuration (3/3)

- SystemVerilog formatting (Verible)
- Linting (Verilator, Verible)
- Code formatting on save
- Syntax highlighting

# 7. Project Structure Setup (1/4)

- Directory Structure
- Source code organization
- Test directory structure
- DUT (Design Under Test) organization
- Configuration files
- `tools/` directory for tools (Verilator as git submodule, UVM as tarball)

# 7. Project Structure Setup (2/4)

- Git Submodules Management
- Verilator is managed as a git submodule in `tools/verilator/`
- UVM is downloaded as a tarball to `tools/uvm-2017/` (not a git submodule)
- Initialize submodules: `./scripts/init_submodules.sh` or `git submodule update --init --recursive`
- Update submodules: `./scripts/update_submodules.sh` or `git submodule update --remote`
- Add new submodule: `./scripts/add_submodule.sh <repo_url> <path>`

# 7. Project Structure Setup (3/4)

- Remove submodule: `./scripts/remove_submodule.sh <path>`
- Makefile/Configuration
- Simple Makefile for running tests
- Verilator configuration
- Environment variable management
- Version Control

# 7. Project Structure Setup (4/4)

- Git initialization
- .gitignore for SystemVerilog and simulation (include build artifacts, waveforms)
- Initial commit structure
- Git submodules in `.gitmodules` file (only Verilator, not UVM)

# 8. First "Hello World" Verification Test (1/3)

- Prerequisites
- Ensure all tools are installed:
  `./scripts/installation.sh`
- Verify tools are accessible: `verilator --version`
  and `ls $UVM_HOME/uvm_pkg.sv` (after sourcing `s...
- Simple DUT Creation
- Basic Verilog module (e.g., AND gate)
- Simple testbench structure

# 8. First "Hello World" Verification Test (2/3)

- SystemVerilog Test
- Basic SystemVerilog testbench structure
- Clock generation
- Signal driving and reading
- Running the test
- UVM Test

# 8. First "Hello World" Verification Test (3/3)

- First UVM test class
- Basic UVM phases
- Running UVM test
- Understanding output

# 9. Verilator Limitations and Workarounds (1/3)

- Known Limitations
- Limited SystemVerilog support
- Some UVM classes may not work
- Complex TLM connections may fail
- Full assertion support limited
- Workarounds

# 9. Verilator Limitations and Workarounds (2/3)

- Use simplified patterns where possible
- Document commercial alternatives
- Focus on concepts over tool features
- Commercial Simulator Alternatives
- When Verilator doesn't support a feature
- How to adapt examples for commercial tools

# 9. Verilator Limitations and Workarounds (3/3)

- Compatibility notes

# 10. Troubleshooting Common Issues (1/4)

- Verilator Issues
- Compilation errors
- Missing dependencies
- Version compatibility
- Path issues
- SystemVerilog syntax errors

# 10. Troubleshooting Common Issues (2/4)

- UVM Library Issues
- Import errors
- Path configuration
- Version compatibility
- Compilation errors
- Build System Issues

# 10. Troubleshooting Common Issues (3/4)

- Makefile errors
- Include path problems
- Compilation flags
- IDE Issues
- SystemVerilog syntax highlighting
- Import resolution problems

# 10. Troubleshooting Common Issues (4/4)

- Debugging not working

# 11. Verification Checklist (1/3)

- C/C++ compiler installed and working
- Verilator installed and verified
- Using script: `./scripts/install_verilator.sh --from-submodule`
- Verify: `verilator --version`
- UVM library installed and verified
- Using script: `./scripts/installation.sh` (UVM is installed automatically)

# 11. Verification Checklist (2/3)

- Verify: `ls $UVM_HOME/uvm_pkg.sv` (after sourcing `scripts/setup_uvm_env.sh`)
- All tools verified together: Run `./scripts/installation.sh` and verify with `verilator --version`...
- IDE configured and working
- First test runs successfully
- Can create and run simple testbench
- Understand basic project structure

# 11. Verification Checklist (3/3)

- Know how to get help when stuck
- Understand Verilator limitations

# Hands-on examples

# Module 0 self-check

```
$ verilator --version
```

# Demo: Verify toolchain

```
$ verilator --version && source scripts/setup_uvm_env.sh 2>/dev/null
      || true && test -f
      "${UVM_HOME:-tools/uvm-2017/1800.2-2017-1.0/src}/uvm_pkg.sv" &&
```

# Demo: Project layout

```
$ ls -la docs/MODULE0.md module1/README.md scripts/installation.sh
```

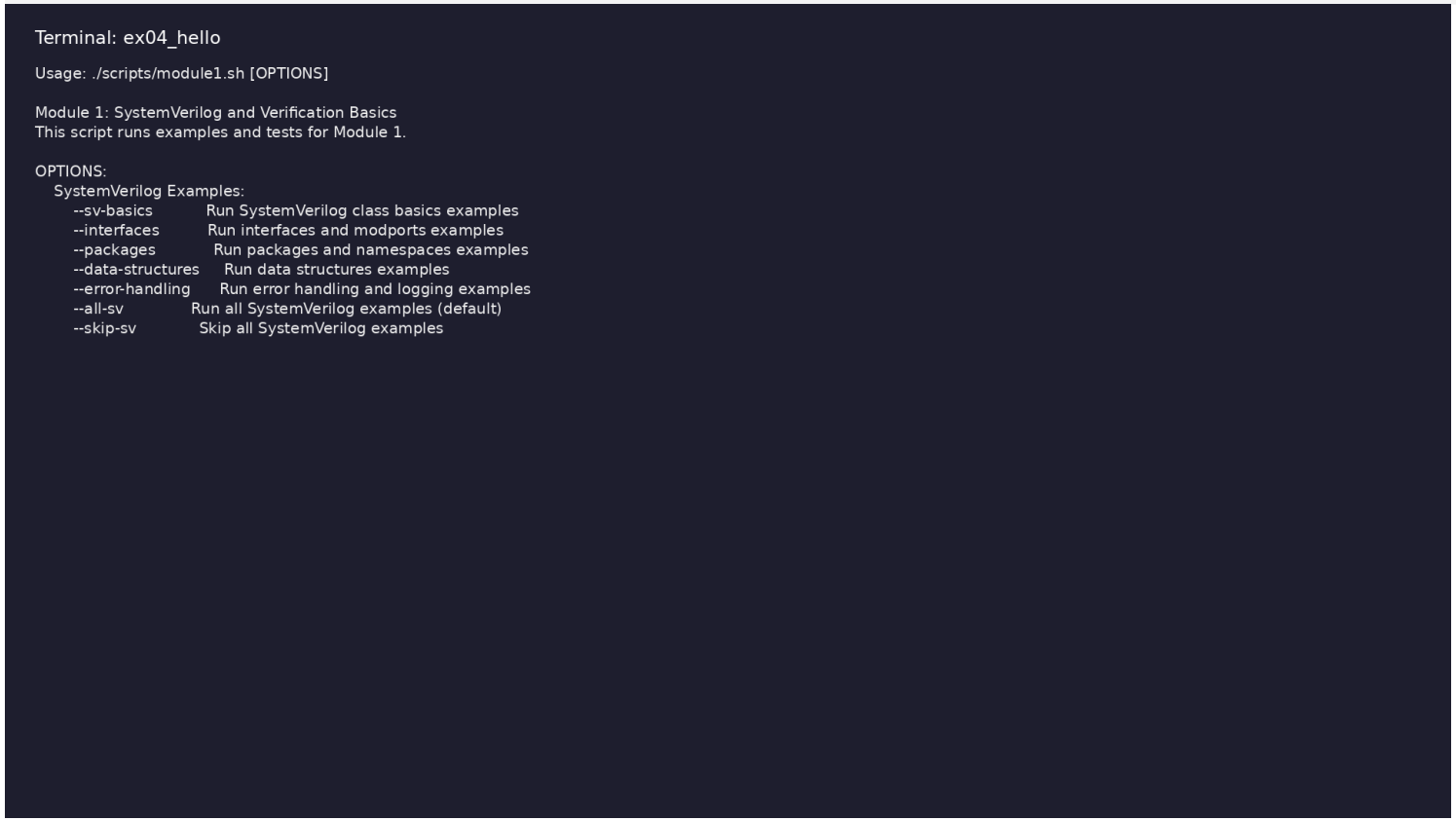
(screenshot pending: assets/screenshots/ex02_project.png)

# Demo: Optional full install

```
$ ./scripts/installation.sh --help | head -20
```

# Demo: First smoke test

```
$ ./scripts/module1.sh --help | head -15
```



# Practice & assessment

# What you should know (1/2)

- Install and configure all required tools
- Set up Verilator
- Install and verify UVM library
- Configure your IDE for verification work
- Create a basic project structure
- Run a simple verification test

# What you should know (2/2)

- Understand Verilator limitations
- Troubleshoot common installation issues

# Exercises

- Installation Verification
- Environment Setup
- First Test
- IDE Configuration
- Project Structure

# Assessment checklist

- Can install all required tools independently
- Can set up Verilator
- Can verify Verilator installation
- Can verify UVM library installation
- Can configure IDE for verification work
- Can create and run a simple test

# Summary & next steps

- Set up development environment and verify installation
- Complete module0/CHECKLIST.md
- Review module0/EXAMPLES.md and run each lab
- Next: Next module in course